Nagar YuwakShikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
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Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department *To*

nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration • Holistic, ethical and sustainable development for lifelong success.*

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.
Hrutik Gujarkar
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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15/07/2026

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Session	**2025-26 (EVEN)**	**Course Name**	**LAB: Optimization Techniques**
Semester	**7th**	**Course Code**	**23ADS1707**
Roll No	**39**	**Name of Student**	**Hrutik Gujarkar**

Practical Number	**04**
Course Outcome	CO1: Understand unconstrained and constrained optimization techniques. CO2: Apply non-linear optimization techniques to solve engineering problems. CO3: Analyze optimization problems using operations research techniques. CO4: Evaluate transportation and transshipment optimization problems.
Aim	To solve a constrained optimization problem using the Lagrange Multipliers Method and determine the optimum value of the objective function.
Problem Definition	Constrained Optimization Using Lagrange Multipliers Method

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Theory (100 words)	The Lagrange Multiplier Method is used to optimize a function subject to one or more equality constraints. Consider the objective function: Subject to: The Lagrangian function is defined as: where: • = Objective Function • = Constraint Function • = Lagrange Multiplier The optimum point is obtained by solving: simultaneously.

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Procedure and Execution (100 Words)	1. Define objective function: f(x,y)=xy 2. Define constraint: x+y−10=0 3. Construct Lagrangian: L=xy+λ(x+y−10) 4. Find partial derivatives: ∂L∂x,∂L∂y,∂L∂λ 5. Set derivatives to zero. 6. Solve the equations. 7. Get optimal values: x=5, y=5 8. Calculate objective: f(5,5)=25 9. Display results: Maximum = 25 at (5,5). .

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Code:

```
import sympy as sp

# Define variables x, y, lam =
sp.symbols('x y lam')

# Objective function f
= x * y

# Constraint function g
= x + y - 10

# Lagrangian Function
L = f + lam * g

# Partial derivatives
eq1 = sp.diff(L, x) eq2
= sp.diff(L, y) eq3 =
sp.diff(L, lam)

# Solve equations solution = sp.solve((eq1, eq2,
eq3), (x, y, lam))

print("Optimal Solution:") print(solution)

# Objective function value
x_val = solution[x] y_val
= solution[y]
```

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```
optimal_value = f.subs({x: x_val, y: y_val})
 print("\nMaximum Value of Objective
Function:") print(optimal_value)
```

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	Output:
	Optimal Solution: **{lam: -5, x: 5, y: 5}** **Maximum Value of Objective Function:** **25**
Output Analysis	The program finds the optimal solution as **x = 5 and y = 5**, with **λ = -5**. The maximum value of the objective function xy is **25**. The constraint x+y=10 is also satisfied. Thus, the **maximum value is 25 at (5,5)**.
Link of student Github profile where lab assignment has been uploaded	-Technique-Practicals
Conclusion	The optimal point is **((5,5))**, where the objective function attains its **minimum value of 50**.
Plag Report (Similarity index < 12%)	
Date	**15/07/2026**